Tao Wen

CONTACT INFORMATION

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RESEARCH INTERESTS

Complexity Science, Network Science, Game Theory, Fractal Property, Group Decision-making

EDUCATION

University of Manchester (UoM) — Manchester, UK
Integrated PhD and MSc in Data Analytics and Society (ESRC funded) — Expect Sep. 2025
Advisor: Prof. Yu-wang Chen and Assistant Prof. Tahir Abbas Syed

Singapore University of Technology and Design (SUTD) — Singapore
MEng. (Research) in Science, Mathematics & Technology — Sep. 2021
Advisor: Assistant Prof. Kang Hao Cheong

Northwestern Polytechnical University (NWPU) — Xi'an, China
B. S. in *Detection, Guidance, and Control Technology (System Engineering)* — June 2019
Advisor: Prof. Wen Jiang and Prof. Yong Deng

SELECTED PUBLICATIONS
[Google Scholar]
Citation: 600+
H-Index: 14
I10-Index: 20

* Corresponding Author; † Contribute equally.

[1] **Tao Wen**, Kang Hao Cheong*, Joel Weijia Lai, Jin Ming Koh, Eugene V. Koonin. "Extending the lifespan of multicellular organisms via periodic or stochastic intercellular competition," *Physical Review Letters*. vol. 128, p. 218101, 2022. (**The Highest Impact Letter in Physics**)

[2] Kang Hao Cheong*†, **Tao Wen**†, Sean Benler, Jin Ming Koh, Eugene V. Koonin*. "Alternating lysis and lysogeny is a winning strategy in bacteriophages due to Parrondo's Paradox," *Proceedings of the National Academy of Sciences*. vol. 119, p. e2115145119, 2022.

[3] **Tao Wen**, Jinde Cao, Kang Hao Cheong*. "Gravity-based Community Vulnerability Evaluation Model in Social Networks: GBCVE," *IEEE Transactions on Cybernetics*. vol. 53, no. 4, pp. 2467-2479, 2023. (Grade 3 journal in ABS list)

[4] **Tao Wen**, Kang Hao Cheong*. "The Fractal Dimension of Complex Networks: A Review," *Information Fusion*. vol. 73, pp. 87-102, 2021. (ESI Highly Cited Paper)

[5] **Tao Wen**, Yong Deng*. "The vulnerability of community structure in complex networks: An entropy approach," *Reliability Engineering & System Safety*. vol. 196, p. 106782, 2020. (ESI Highly Cited Paper, Grade 3 journal in ABS list)

[6] **Tao Wen**, Yong Deng*. "Identification of influencers in complex network by local information dimensionality," *Information Sciences*. vol. 512, pp. 549-562, 2020. (ESI Highly Cited Paper)

[7] Qiuya Gao, **Tao Wen**, Yong Deng*. "Information Volume Fractal Dimension". *Fractals*. vol. 29, p. 2150263, 2021. (ESI Hot Paper & ESI Highly Cited Paper)

RESEARCH EXPERIENCE

Data Analytics for Addressing Fake News and Deepfakes in Social Networks — Sep. 2021–Now
University of Manchester, Advisor: Professor Yu-wang Chen and Assistant Prof. Tahir Abbas Syed

- Analyzed communication structure and flow within an organizational hierarchy through the perspective of social network analysis.
-

Quantifying Impact of Social Influence on Language Dynamics — Feb.–Mar. 2023
University of Oxford **(Funded by Turing Exchange Fellowship)**, Advisor: Professor Renaud Lambiotte

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Studying Economic Growth and Environmental Sustainability through Cooperation May–Sep. 2022
University of Manchester, Advisor: Professor Yu-wang Chen and Dr. Ting Wu

▪

Analyzing Parrondo's Paradox Phenomenon in Complex Networks Jan. 2020–Sep. 2021
Singapore University of Technology and Design, Advisor: Assistant Professor Kang Hao Cheong

- Explored the counterintuitive phenomena in (a) the evolution of bacteriophages, (b) the intercellular competition in multicellular organisms, (c) the prey dormancy of predator-prey systems, and (d) the implementation of different strategies under the epidemic of COVID-19 – attributed to Parrondo's paradox.
- Reviewed the development and progress of fractal characteristics in the network systematically, including network covering, various concepts of dimensions, and their real-world applications.

Finding Important Properties of Nodes and Communities in Complex Networks June–Sep. 2019
University of Electronic Science and Technology of China, Advisor: Professor Yong Deng

- Combined the internal and external factors of communities to measure each community's vulnerability and improved the recognition accuracy in real-world network applications via the entropy-based method.
- Collected the personal opinion of each expert by 2-order additive fuzzy measure and identified the most important expert after constructing the expert network based on their relationships, thereby solving group decision-making problems without sufficient information.

Exploring the Properties of Networks by Different Dimensions May 2016–June 2019
Northwestern Polytechnical University, Advisor: Professor Wen Jiang & Yong Deng

- Promoted information dimension and Rényi dimension into weighted complex networks and explored complex networks' fractal and self-similarity properties.
- Developed several recognition models based on the dimension of the network to solve practical problems, such as measuring the complexity and vulnerability of airline networks and identifying the similarity and importance of individuals.

Evaluating Vulnerability of Networks Based on Fuzzy Fractal Dimension May 2017–May 2018
Team Leader, National Undergraduate Training Programs for Innovation and Entrepreneurship (￥20, 000)

An Autonomous Landing Scheme for Cargo Drone Based on Computer Vision May 2017–May 2018
Researcher, National Undergraduate Training Programs for Innovation and Entrepreneurship (￥20, 000)

Exploring the Composition and Future Scalability of Smart House Jan. 2019–Feb. 2019
Australian National University, Summer Session Course Project

AWARDS & HONORS

- **Turing Exchange Fellowship** 2022
- Excellent paper award of International Postgraduate Interdisciplinary Forum on Smart Shipping and Green Future 2022
- Decision and Cognitive Sciences Research Centre Support Fund 2022
- ESRC funded integrated PhD and MSc in Data Analytics and Society Scholarship 2021
- Singapore University of Technology and Design MEng (Research) Fellowship 2020
- The "Challenge Cup" National Undergraduate Extracurricular Academic Science and Technology Contest: **Special First Prize (Top 4%)** 2019
  Highest award in the field of information science in natural science papers.
  The "Challenge Cup" is the **highest academic science and technology competition** in China.
- The "Challenge Cup" National Undergraduate Extracurricular Academic Science and Technology Contest in Shaanxi Area: **Outstanding Winner (TOP 5%)** 2019
  Best grade in the field of natural science papers in NWPU **so far**.

- **Outstanding Student Pacemaker in Northwestern Polytechnical University (TOP 0.1%)** 2018
  Only **10 undergraduates** are awarded among more than 12,000 undergraduates in NWPU.
- **China National Scholarship (Top 1%)** 2018
  Highest honor for students' year achievement in the School of Electronics and Information.
- Global Mathematical Contest in Modeling (MCM/ICM): **Meritorious Winner** 2018
- Chinese Undergraduate Mathematical Contest in Modeling in Shaanxi Area: **First Prize** 2017
- Second Prize scholarship of AVIC Optronics Institute (**TOP 5%**) 2017
- First Prize scholarship of Northwestern Polytechnical University (**TOP 10%**) 2017
- Global Mathematical Contest in Modeling (MCM/ICM): **Honorable Mention** 2017

ACADEMIC SERVICE

- Presentation at International Postgraduate Interdisciplinary Forum on Smart Shipping and Green Future in Shanghai Jiao Tong University (Online), Shanghai, China, 2022.
- Special Issue Editor on *Mathematics* (Data-driven Decision Making: Models, Methods and Applications), 2022.
- Teaching Assistant of BMAN73701, Programming in Python for Business Analytics, UoM, 2022.
- Conference presentation at the 26th International Conference on Multiple Criteria Decision Making, Portsmouth, UK, 2022.
- Talk at the Oberseminar Dynamics Seminar at the Technical University of Munich (Online), Munich, Germany, 2021.
- Teaching Assistant of 10.018, Modelling Space and Systems, SUTD, 2020.
- Conference presentations at the 9th China Information Fusion Conference, Taiyuan, China, 2019.
- Talk at Ao Xiang Students' Lecture in NWPU, 2019.
- Independent reviewer for:
  - *Applied Sciences*
  - *Bioengineering*
  - *Communications Physics*
  - *Chaos: An Interdisciplinary Journal of Nonlinear Science*
  - *Electronics*
  - *Humanities & Social Sciences Communications*
  - *IEEE Transactions on Signal and Information Processing over Networks*
  - *International Journal of Intelligent Systems*
  - *Journal of Computational Science*
  - *Mathematics*
  - *Mathematical Biosciences and Engineering*
  - *Physica A: Statistical Mechanics and Its Applications*
  - *Reliability Engineering & System Safety*
  - *Scientific Reports*
  - *Sensors*

SOCIAL ACTIVITIES

- **Volunteer** | International Conference on Intelligent Unmanned Systems 2018
- **Volunteer** | Northwestern Polytechnical University 80th Anniversary 2018

SKILLS

Technology: Python, MATLAB, R language, C language, LaTeX, Gephi, Origin, UCINET, MPNet.
Languages: Mandarin, English.

REFERENCES

Prof. Dr. Yu-Wang Chen
Alliance Manchester Business School

Prof. Dr. Yong Deng
Institute of Fundamental and Frontier Science

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Manchester, UK
✉ Yu-wang.Chen@manchester.ac.uk

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Chengdu, Sichuan, P. R. China
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Assistant Prof. Dr. Kang Hao Cheong
Science, Mathematics & Technology Cluster
Singapore University of Technology and Design
Singapore
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Prof. Dr. Wen Jiang
School of Electronics and Information
Northwestern Polytechnical University
Xi'an, Shaanxi, P. R. China
✉ jiangwen@nwpu.edu.cn

FULL PUBLICATIONS LIST

2022

[1] **Tao Wen**, Jinde Cao, Kang Hao Cheong*. "Gravity-based Community Vulnerability Evaluation Model in Social Networks: GBCVE," *IEEE Transactions on Cybernetics*. vol. 53, no. 4, pp. 2467-2479, 2023. (Grade 3 journal in ABS list)

2022

[2] **Tao Wen**, Kang Hao Cheong*, Joel Weijia Lai, Jin Ming Koh, Eugene V. Koonin. "Extending the lifespan of multicellular organisms via periodic or stochastic intercellular competition," *Physical Review Letters*. vol. 128, p. 218101, 2022. (**The Highest Impact Letter in Physics**)

[3] Kang Hao Cheong*†, **Tao Wen**†, Sean Benler, Jin Ming Koh, Eugene V. Koonin*. "Alternating lysis and lysogeny is a winning strategy in bacteriophages due to Parrondo's Paradox," *Proceedings of the National Academy of Sciences*. vol. 119, p. e2115145119, 2022.

[4] **Tao Wen**, Qiuya Gao, Yu-wang Chen, Kang Hao Cheong*. "Exploring the Vulnerability of Logistics Networks by Entropy: An Asia-Europe Maritime Transportation Network Case Study," *Reliability Engineering & System Safety*. vol. 226, p. 108578, 2022. (Grade 3 journal in ABS list)

[5] **Tao Wen**, Qiuya Gao, Tamás Kalmár-Nagy, Yong Deng*, Kang Hao Cheong*. "A Review of Predator-Prey Systems with Dormancy of Predators," *Nonlinear Dynamics*. vol. 107, pp. 3271-3289, 2022.

[6] **Tao Wen**, Joel Weijia Lai, Kang Hao Cheong*. "Switching between two losing stocks may enable paradoxical win: an empirical analysis," *Fractals*. Accepted.

[7] **Tao Wen**, Huiling Chen, Kang Hao Cheong*. "Visibility Graph for Time Series Prediction and Image Classification: A Review," *Nonlinear Dynamics*. vol. 110, pp. 2979-2999, 2022.

[8] Jie Zhao, **Tao Wen**, Hadi Jahanshahi, Kang Hao Cheong*. "The random walk-based gravity model to identify influential nodes in complex networks," *Information Sciences*. vol. 609, pp. 1706-1720, 2022.

[9] Qiuya Gao, **Tao Wen**, Yong Deng*. "A Novel Network-based and Divergence-based Time Series Forecasting Method," *Information Sciences*. vol. 612, pp. 553-562, 2022.

2021

[10] **Tao Wen**, Kang Hao Cheong*. "The Fractal Dimension of Complex Networks: A Review," *Information Fusion*. vol. 73, pp. 87-102, 2021. (**ESI Highly Cited Paper**)

[11] **Tao Wen**, Eugene V. Koonin, Kang Hao Cheong*. "An alternating active-dormitive strategy enables disadvantaged prey to outcompete the perennially active prey through Parrondo's paradox," *BMC Biology*. vol. 19, p. 168, 2021.

[12] Qiuya Gao, **Tao Wen**, Yong Deng*. "Information Volume Fractal Dimension". *Fractals*. vol. 29, p. 2150263, 2021. (**ESI Hot Paper & ESI Highly Cited Paper**)

[13] Yayun Dai, Guangjie Zhan, Ye Ye, Wei Bao, **Tao Wen**, Kang Hao Cheong*, Nenggang Xie*. "Game dynamics of emotion evolution based on the Moran process," *Chaos: An Interdisciplinary Journal of Nonlinear Science*. vol. 31, p. 033153, 2021.

[14] Prabal Datta Barua, Nadia Fareeda Muhammad Gowdh, Kartini Rahmat, Norlisah Ramli, Wei Lin Ng, Wai Yee Chan, Mutlu Kuluozturk, Sengul Dogan*, Mehmet Baygin, Orhan Yaman, Turker Tuncer, **Tao Wen**, Kang Hao Cheong*, U. Rajendra Acharya. "Automatic COVID-19 detection using exemplar hybrid deep features with X-ray images," *International Journal of Environmental Research and Public*

Health. vol. 18, p. 8052, 2021.

2020

[15] Kang Hao Cheong*, **Tao Wen**, Joel Weijia Lai. "Relieving cost of epidemic by Parrondo's paradox: A COVID-19 case study," *Advanced Science*. vol. 7, p. 2002324, 2020. [Press]

[16] **Tao Wen**, Danilo Pelus, Yong Deng*. "Vital Spreaders Identification in Complex Networks with Multi-Local Dimension," *Knowledge-Based Systems*. vol. 195, p. 105717, 2020.

[17] **Tao Wen**, Yong Deng*. "The vulnerability of community structure in complex networks: An entropy approach," *Reliability Engineering & System Safety*. vol. 196, p. 106782, 2020. (**ESI Highly Cited Paper**, Grade 3 journal in ABS list)

[18] **Tao Wen**, Yong Deng*. "Identification of influencers in complex network by local information dimensionality," *Information Sciences*. vol. 512, pp. 549-562, 2020. (**ESI Highly Cited Paper**)

2019

[19] **Tao Wen**, Shuyu Duan, Wen Jiang*. "Node similarity measuring in complex networks with relative entropy," *Communications in Nonlinear Science and Numerical Simulation*. vol. 78, p. 104867, 2019.

[20] **Tao Wen**, Wen Jiang*. "Identifying influential nodes based on fuzzy local dimension in complex networks," *Chaos, Solitons & Fractals*, vol. 119, pp. 332-342, 2019.

[21] **Tao Wen**, Wen Jiang*. "Measuring the complexity of complex network by Tsallis entropy," *Physica A: Statistical Mechanics and Its Applications*. vol. 526, p. 121054, 2019.

[22] Shuyu Duan, **Tao Wen**, Wen Jiang*. "A new information dimension of complex network based on Rényi entropy," *Physica A: Statistical Mechanics and Its Applications*, vol. 516, pp. 529–542, 2019.

2018

[23] **Tao Wen**, Moxian Song, Wen Jiang*. "Evaluating topological vulnerability based on fuzzy fractal dimension," *International Journal of Fuzzy Systems*, vol. 20, pp. 1956–1967, 2018.

[24] **Tao Wen**, Wen Jiang*. "An information dimension of weighted complex networks," *Physica A: Statistical Mechanics and Its Applications*, vol. 501, pp. 388 – 399, 2018.

Conference paper

[25] **Tao Wen**, Yu-wang Chen*, Tahir Abbas Syed. "Analyzing Communication Patterns and Dynamics in an Organizational Hierarchy through Social Network Analysis," *Advances in Data Science and AI Conference 2023*. Manchester, UK, June 2023. (Abstract)

[26] **TBD**. "Analysing communication network structure and dynamics in an organizational hierarchy," *23rd Conference of the International Federation of Operational Research Societies*. Santiago, Chile, July 2023. (Abstract)

[27] **Tao Wen**, Yu-wang Chen*, Tahir Abbas Syed. "Ranking Influential Users in Social Networks by the Evidential Reasoning Algorithm," *26th International Conference on Multiple Criteria Decision Making (MCDM2022)*. Portsmouth, UK, June 2022. (Abstract)

[28] **Tao Wen**, Shuyu Duan, Wen Jiang*. "Forecasting time series based on visibility graph and relative entropy," *The Ninth Chinese Information Fusion Conference*, Taiyuan, China, Oct. 2019. (In Chinese)

[29] Shuyu Duan, **Tao Wen**, Xinyang Deng, Wen Jiang*. "Identifying influential nodes based on Tsallis entropy and information dimension," *The Ninth Chinese Information Fusion Conference*, Taiyuan, China, Oct. 2019. (In Chinese)

[30] Shuai Xu, Zichang He, **Tao Wen**, Wen Jiang*. "A Physarum-inspired Model for the Path Planning of Uninhabited Combat Air Vehicle," *The Eighth Chinese Information Fusion Conference*, Xi'an, China, July 2018. (In Chinese)

Paper submitted to Journal

[31] **Tao Wen**, Eugene V. Koonin, Kang Hao Cheong*. "Bending the Rules of Coevolution: How Flexible Parasites Can Outsmart Their Hosts for Evolutionary Dominance," *TBD*.

[32] **Tao Wen**, Yu-wang Chen*, Tahir Abbas Syed, Darminder Ghataoura. "Examining communication network structure and dynamics in an organizational hierarchy: A social network analysis approach," *TBD*.

[33] **Tao Wen**, Yu-wang Chen*, Tahir Abbas Syed, Ting Wu. "ERIUE: Evidential Reasoning-based Influential Users Evaluation in Social Networks," *TBD*.

[34] Zeyi Liu†, **Tao Wen**†, Yong Deng*, Hamido Fujita. "Cooperation-guided Experts Importance Identification Model with Fuzzy Framework: A Network Design," *TBD*.

[35] Ankit Mishra, **Tao Wen**, Kang Hao Cheong*. "Enhancing Network Communication with Brownian-Inspired Game Theory," *TBD*.

[36] Qiuya Gao, **Tao Wen**, Xiangfeng Dai, Mengge Chai, Kang Hao Cheong*, Zhen Wang*. "Measuring Similar Nodes in Complex Networks based on Gravity Model," *TBD*.

[37] Kang Hao Cheong*, Michael C. Jones, **Tao Wen**. "'Governing' the ungovernable: When will we learn?" *TBD*.

[38] Chao Wang, Songjie Li, Miao Shi, Jie Zhao, **Tao Wen**, U. Rajendra Acharya, Neng-gang Xie*, Kang Hao Cheong*. "Multi-objective squirrel search algorithm for EEG feature selection," *TBD*.

Last updated: May 5, 2023